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Simulations of biological associative learning behaviors using Cs_2SnI_6 perovskite memristors

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ABSTRACT

The development of artificial intelligence devices has received a lot of attention. In this study, Cs_2SnI_6 perovskite memristors with Ag/Cs_2SnI_6/indium tin oxide sandwich structure were prepared, exhibiting bipolar resistive switching behaviors, which was used for simulations of biological associative learning behaviors. The short-term plasticity of biological synapses was mimicked and the conductance changes under stimulation with pulsed signals of different intervals and voltage amplitudes were explored. The device also enabled the simulation of long-term plasticity of biological synapses, including long-term potentiation and long-term depression. Moreover, the device remained excellently stable under 1500 consecutive voltage pulses. In addition, the cognitive processes of "learning-forgetting-relearning" in the human brain and the proboscis extension response behavior of bees were also simulated by the present devices, demonstrating its great potential to simulate the biological associative learning behaviors.

1. Introduction

The development of the information age has brought about a surge in the amount of data and information, which requires higher efficiency of computers in data acquisition, encoding and processing [1]. Due to the physical isolation of computing modules and storage units in traditional digital computer systems, the current systems based on the von Neumann architecture have encountered a bottleneck in technological development [2,3]. To break through this limitation, inspired by the highly parallel data processing capability of human brain, neuromorphic computing systems [4,5], that is, the utilization of electronics to simulate the functions of the biological brain, have been proposed to achieve more efficient and low-power information processing. Among them, artificial synapses are considered to be the basic hardware components for building brain-like computing systems [6,7]. Thus, the investigation of learning and memory mechanisms is crucial for artificial synapse research. Associative learning behavior, i.e., weighted connections between neurons formed by the joint action of biological principles and practical experience, is a key concept in neuromorphic computing [8]. Classical conditioned reflexes, as a basic form of associative learning, have the characteristics of acquisitiveness, extinction, and generalization, which correspond to the functions of information storage, old information elimination, and new information storage of artificial synaptic devices, respectively [9]. Therefore, synaptic devices with associative learning function would have the great potential applications in the fields of brain-computer interface, neural control and medical health [10].

Memristor is one of ideal devices for hardware implementation of artificial synapses, capable of simulating the basic functions of biological synapses. Memristors is a sandwich structure device, which consist of top electrode, functional layer and bottom electrode, corresponding to the presynaptic membrane, the synaptic gap and the postsynaptic membrane in the synapse [11], respectively. Until now, the fast switching speed, low power consumption, and high-density data storage capability of memristors have attracted extensive attention in the fields of data storage and artificial neural networks [12,13]. The functional layer materials of memristors generally include metal oxides [14,15], metal sulphides [16], two-dimensional materials [17,18], organic materials [19], and perovskite materials [20–22]. By utilizing TaO/lGZO as the functional layer, Li et al. proposed perception-memory integrated system, which gives it tremendous potential in the fields of disability, human–computer interaction, and simulation of the human retina and brain [5]. The bioinspired inotropic synapse fibers was reported for ultralow-power multiplexing neuromorphic sensorimotor textiles [7].

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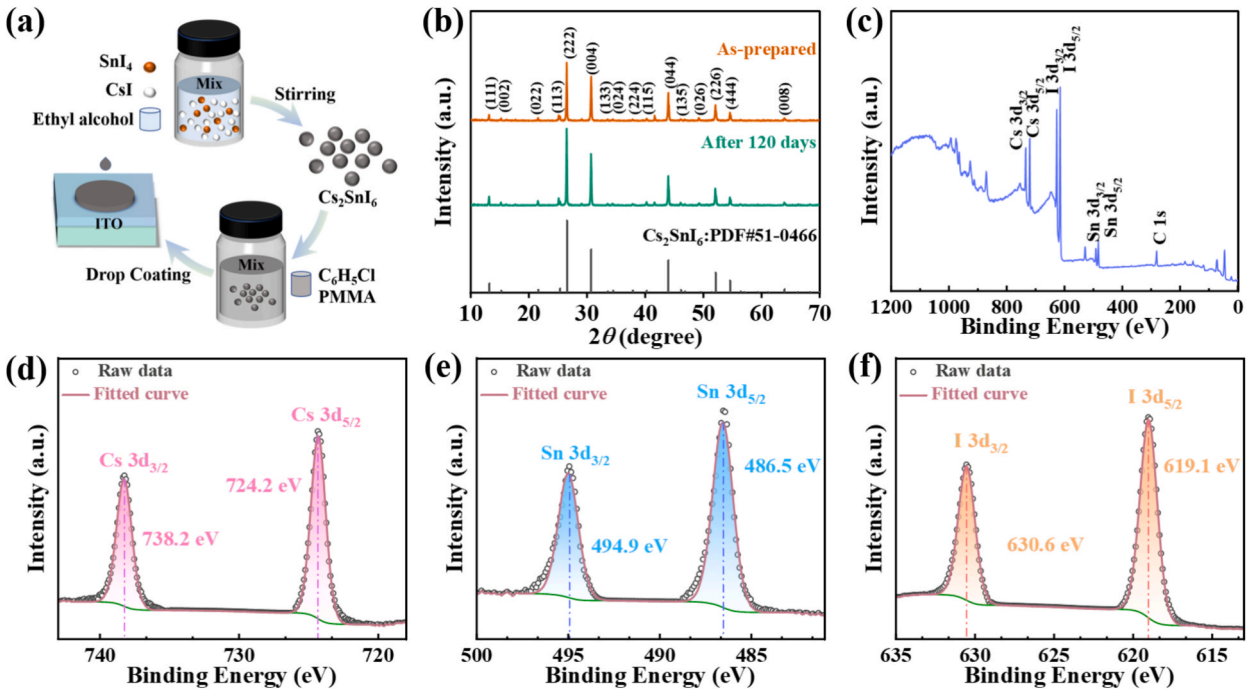


Fig. 1. (a) Sample synthesis and device preparation. (b) XRD pattern of Cs_2SnI_6 powder. (c) Full-range XPS spectrum of the Cs_2SnI_6 film and high-resolution XPS spectra of (d) Cs 3d (e) Sn 3d and (f) I 3d.

Meanwhile, two-dimensional materials were applied to optoelectronic components [23], biosensors [24] and high-speed non-volatile flash memory device with the atomically engineered scale [25]. In recent years, perovskite materials have been widely used as the function layer of memristors due to their tunable band gap [26–28], long carrier diffusion length, and strong light absorption [27]. However, the toxicity and instability of traditional lead-based perovskite materials have become the main limiting factors for practical applications [29]. It is urgent to seek for lead-free perovskite materials acting as the functional layer in memristor [1,29]. Dutta and the co-workers fabricated the Cs-Cu-l perovskite memristor, which paves the way for improved lead-free perovskite-based neuromorphic devices for biological applications [30]. As a lead-free perovskite material, Cs_2SnI_6 films have been explored in various devices, such as solar cells [31], photodetectors [32], self-powered nanogenerators [33] and so on, thanks to its structural stability under air, heat and light conditions [34,35]. Nevertheless, up to date, the study of Cs_2SnI_6 perovskite memristor is rare for simulations of biological associative learning behaviors.

In this study, Cs_2SnI_6 perovskite powder was synthesized by a solution method to prepare Cs_2SnI_6 films using a simple drop-coating technique. Using the thin films, the memristors with Ag/Cs_2SnI_6/indium tin oxide (ITO) sandwich structure were fabricated, showing considerable potential for simulating synaptic functions and associative learning memory functions. Basic synaptic functions such as short-term plasticity and long-term plasticity, including paired-pulse facilitation/depression (PPF/PPD), long-term potentiation (LTP), and long-term depression (LTD) were simulated by using voltage stimulation-induced conductance changes. The conductance triggered by 10 consecutive pulses (0.3 V, 10 ms) was significantly increased compared to the resting conductance, indicating that the Ag/Cs_2SnI_6/ITO device can realize the transition from short-term memory (STM) to long-term memory (LTM). Moreover, the device remained stable under 1500 consecutive pulse stimuli, exhibiting good durability. The "learning-forgetting-relearning" process of the human brain was simulated by applying different voltage pulses to the device. The Ag/Cs_2SnI_6/ITO device has also successfully established the associative learning behaviors, i.e., acquisition, extinction, and generalization of classical conditioned reflexes, by simulating the proboscis extension response (PER) behavior of bees. The present Ag/Cs_2SnI_6/ITO devices show great promise for future applications in the simulation of biological synapses.

2. Experiment and measurements

2.1. Experimental fabrications

The lead-free perovskite Cs_2SnI_6 powder was synthesized by a solution method, as shown in Fig. 1(a). In the process, 0.6 mmol of cesium iodide (CsI) and 0.3 mmol of tin tetraiodide (SnI_4) were dissolved in 1 mL of anhydrous ethanol. Then, the mixture was stirred thoroughly at room temperature for 4 h. After centrifuging and drying at 75 °C/1.5 h, a black precipitate of Cs_2SnI_6 powder was obtained. To grow the Cs_2SnI_6 film, the ITO-glass substrate was firstly cleaned using acetone, ethanol and deionized water sequentially for 15 min. Then, 20 mg of Cs_2SnI_6 powder was ultrasonically mixed with 0.2 ml of PMMA-chloroform solution at a concentration of 5 mg/mL, and it was coated on the ITO substrate using the drop-coating method. The drop-casting process in this work is static, the film is surrounded with solvent vapor while drying, which can protect the drying film from water invasion [36]. To control the thickness and improve the film quality to ensure the denseness of film, the above coated substrate was annealed at 100 °C for 20 min. Finally, Ag as the top electrode was deposited on the functional layer by a magnetron sputtering method.

2.2. Experimental setups

The crystal structures of Cs_2SnI_6 powder were measured using X-ray diffractometer (XRD) with Cu Kα radiation, provided by Japan Rigaku X-ray Corporation. The current–voltage (I-V) characteristics of device were analyzed using a Keithley 2400 instrument, and artificial synaptic behaviors were measured by Keithley 2614B source/meter unit at room temperature. The scanning electron microscopy (SEM) from JEOL JEM-7001F was used to evaluate the morphologies of Cs_2SnI_6 perovskite films. The chemical composition of films was examined by X-ray photoelectron spectroscopy (XPS) from Shimazu-Kratos Analytical (UK)

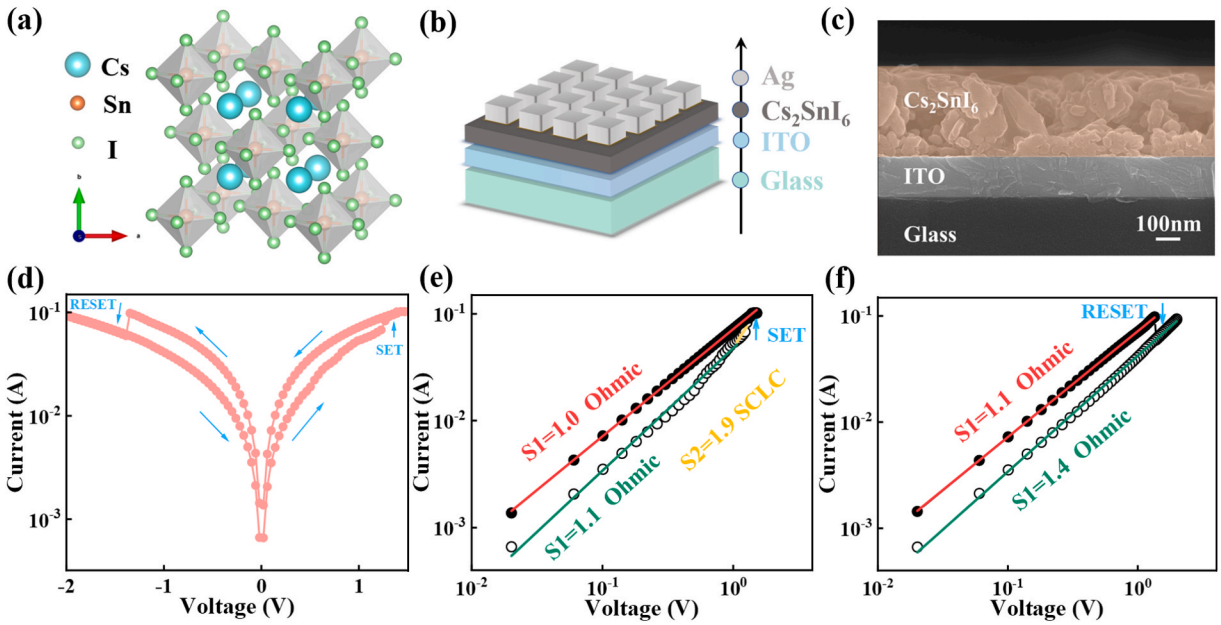


Fig. 2. (a) Illustration of Cs_2SnI_6 crystal structure. (b) Schematic structure of Ag/Cs_2SnI_6/ITO device. (c) The cross-sectional SEM image of Cs_2SnI_6 film grown on ITO glasses. (d) I-V characteristics of Ag/Cs_2SnI_6/ITO. Double logarithmic plots of the I-V characteristics of Ag/Cs_2SnI_6/ITO device during (e) "SET" process and (f) "RESET" process.

using the Model Axis Supra system.

3. Results and discussion

To evaluate the crystallization of Cs_2SnI_6 perovskite films, the XRD measurements were performed, as shown in Fig. 1(b). Five dominant peaks were clearly observed at 26.49°, 30.68°, 43.95°, 52.06°, and 54.56°, corresponding to the (222), (004), (044), (226), and (444) crystallographic planes of as-prepared Cs_2SnI_6, respectively. All diffraction peaks align well with the standard card (No. 51-0466), indicating a cubic crystal structure. No peaks of CsI and SnI_4 were observed, revealing that Cs_2SnI_6 was successfully synthesized without other impurities. Additionally, XRD pattern of Cs_2SnI_6 powder after 120 days was also measured. The diffraction peaks agree well with the as-prepared sample, demonstrating the powder performance remains stable within 120 days. In addition, the qualitative analysis of elemental composition is vital to evaluate a material [37]. Fig. 1(c) is the full XPS spectra of Cs_2SnI_6, indicating it is composed of Cs, Sn and I elements. The peaks at 724.2 eV and 738.2 eV are attributed to Cs $3d_{5/2}$ and Cs $3d_{3/2}$, respectively, as shown in Fig. 1(d). The peaks in Fig. 1(e) at 486.5 eV and 494.9 eV correspond to Sn $3d_{3/2}$ and Sn $3d_{5/2}$, respectively, ensuring that Sn is +4 valence state. The binding energy of I 3d in Fig. 1(f) shows two peaks of I $3d_{5/2}$ and I $3d_{3/2}$ located at 619.1 eV and 630.6 eV, respectively. The present XPS results are consistent with previous reports [38]. Based on the measurements of XRD and XPS, it is concluded that Cs_2SnI_6 has been synthesized successfully. The elemental composition for the present Cs_2SnI_6 film was to also checked using the XPS. It is found that the atomic ratio of Cs, Sn and I was 1.98: 1.0: 5.32 in the Cs_2SnI_6 layer, which is far from the stoichiometric ratio of 2.0: 1.0: 6.0, indicating that a large number of I vacancies existed in the present Cs_2SnI_6 layer [39].

Fig. 2(a) illustrates the Cs_2SnI_6 crystal structure. The Sn atoms are surrounded by six I atoms to form octahedra, where Cs_2SnI_6 consists of isolated $[SnI_6]^{2-}$ clusters that are surrounded by Cs cations. Fig. 2(b) shows the memristor configuration with an Ag/Cs_2SnI_6/ITO structure. The measurement of SEM was performed to investigate the morphology of Cs_2SnI_6 film, as shown in Fig. S1. It is observed that the Cs_2SnI_6 film featured a uniform coverage on the substrate surface. The cross-sectional SEM image of Cs_2SnI_6 film was shown in Fig. 2(c), indicating the thickness of perovskite layer was around 350 nm. The results are comparable to that of the $Cs_3Sb_2Cl_3I_6$ film [40]. Based on the above optimized experimental results, the current Cs_2SnI_6 film is suitable for the fabrication of advanced functional device. The resistance switching (RS) performance of this device was investigated by analyzing the I-V semi-logarithmic curve, as shown in Fig. 2(d). It's worth noting that the device exhibited a bipolar resistive switching behavior by applying scanning voltage of 0 → 1.5 → 0 → -2 → 0 V. The blue arrows denote the sweep directions of applied voltage. The present device exhibits analog RS behavior, rather than digital RS characteristic, which is more suitable for synaptic functions [41–43]. When the voltage changed from 0 to 1.5 V, the device remained a low current in the initial stage, corresponding to a high resistive state (HRS). The electronic current slightly increased at around 1.23 V inducing a low resistive state (LRS) or on-state, which is generally named as the "SET" process, and 1.23 V is defined as V_{SET}. The "RESET" procedure appeared under a negative scanning voltage from 0 to -2 V. The device remained in the LRS until at a sufficient reverse bias of -1.35 V(V_{RESET}) and then a resistance switched from the LRS to the HRS, corresponding to the off-state. To analyze the conduction mechanism of the Ag/Cs_2SnI_6/ITO device, the I-V characteristic curves were replotted on a double logarithmic scale. Fig. 2(e) shows the I-V curve of the "SET" process of the Ag/Cs_2SnI_6/ITO device. During the low voltage range of the positive voltage sweep, the slope of the I-V curve in the HRS region is 1.1, indicating that the device behavior conforms to ohmic conduction ($I \propto V$) [44]. This means that the number of carriers produced by thermal excitation was larger than the number of injected carriers. As a result, the traps were partially filled [45]. As the voltage increases, the slope of the curve rises to 1.9, which is approximately equal to 2. This indicates the device enters the trap-filling limiting region [45,46], following the space charge limiting current (SCLC) model ($I \propto V^2$). The slope of the LRS region is 1.0, which belongs to ohmic conduction, indicating the formation of a conductive path. Fig. 2(f) shows the I-V curve of the "RESET" process of the Ag/Cs_2SnI_6/ITO device. During the negative voltage sweep stage, the current–voltage fitting curve indicates that the HRS with a slope of 1.4 and the LRS with a slope of 1.1 are in line with a linear relationship, corresponding to the ohmic conductance model. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

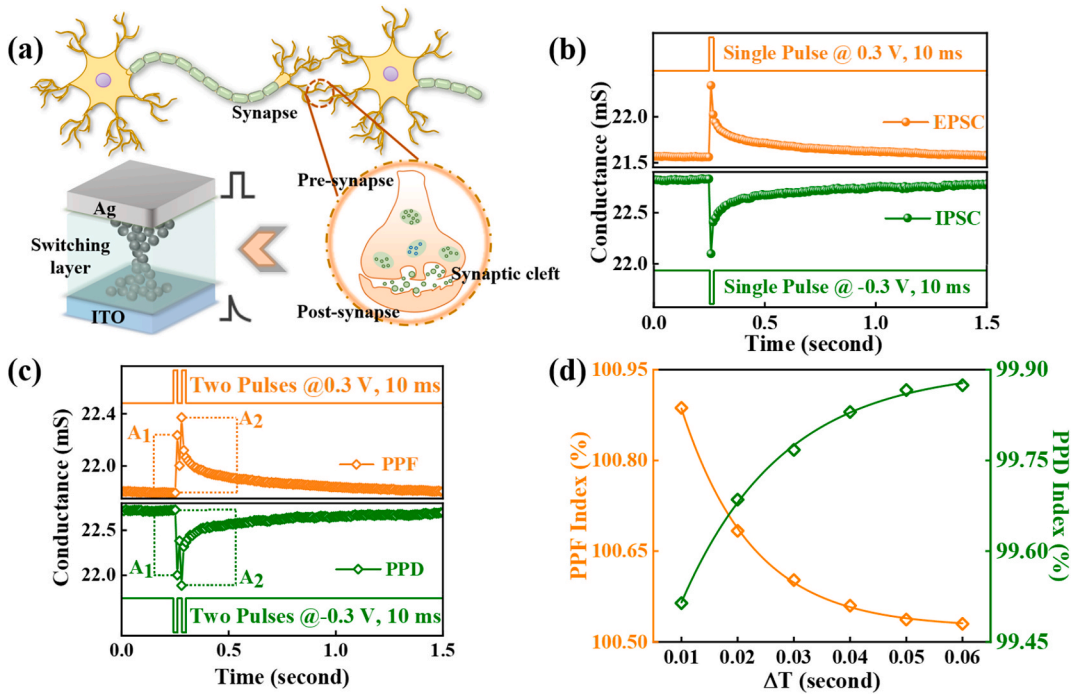


Fig. 3. (a) Schematic diagram of biological synapses in relation to the structure of Ag/Cs_2SnI_6/ITO devices. (b) Conductance response of Ag/Cs_2SnI_6/ITO device under single pulse stimulation (V_{read} = 0.1 V) for EPSC and IPSC. (c) Conductance response of Ag/Cs_2SnI_6/ITO device under double pulse stimulation (V_{read} = 0.1 V) for PPF and PPD. (d) Dependence of PPF (orange rhombus) and PPD (green rhombus) index on ΔT for Ag/Cs_2SnI_6/ITO device. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

The changes in the conductive states of the HRS and LRS in the Ag/Cs_2SnI_6/ITO device can be used to simulate the function of a biological synapse. Fig. 3(a) illustrates the relationship between the structure of biological synapse and the Cs_2SnI_6-based artificial synapse. The Ag electrode, the Cs_2SnI_6 functional layer, and the ITO electrode are corresponding to the presynaptic membrane, the synaptic cleft, and the postsynaptic membrane, respectively. A pulse voltage applied to the Ag electrode causes the migration of halogen vacancies in the perovskite layer, resulting in a change in the device conductivity, which is similar to the change in the strength of the connection between the presynaptic membrane and the postsynaptic membrane. Therefore, this device has the potential to simulate the biological synapses, where synaptic plasticity refers to the property that the strength of synaptic connections can be continuously modulated, including short-term plasticity and long-

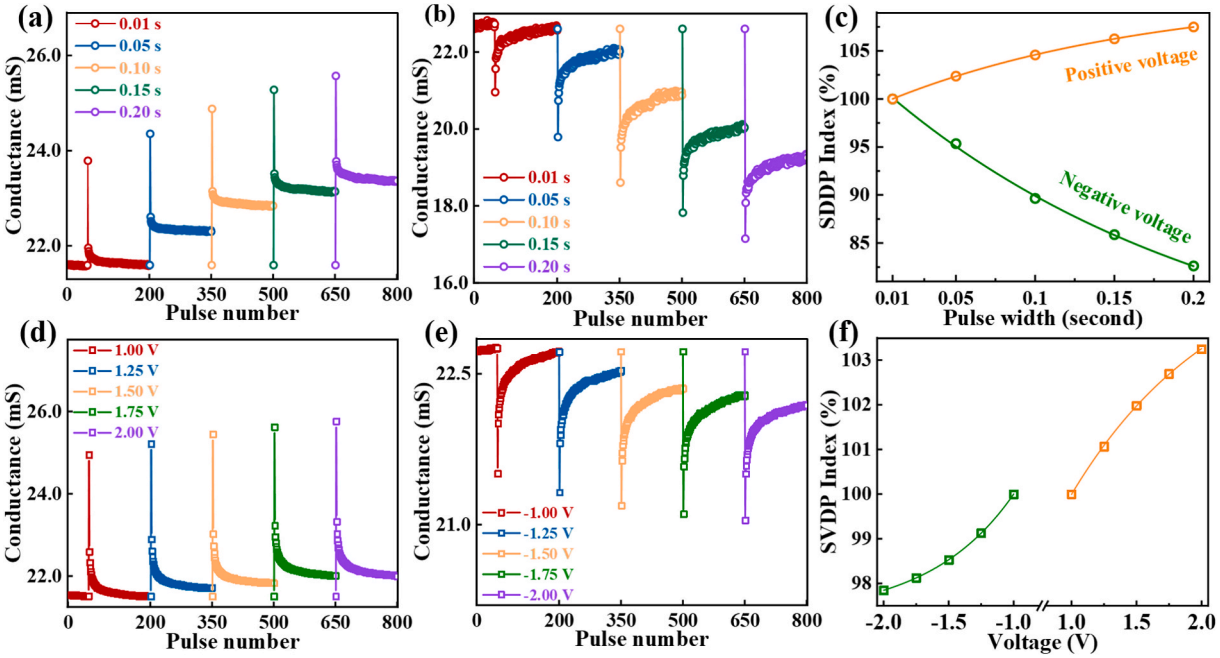


Fig. 4. Conductance changes in the Ag/Cs_2SnI_6/ITO device stimulated with a pulse voltage of 0.3 V (a)/-0.3 V (b) under different pulse widths. (c) SDDP index versus pulse width in the Ag/Cs_2SnI_6/ITO device under the positive (0.3 V) and negative (-0.3 V) voltages. Conductance changes in the Ag/Cs_2SnI_6/ITO device induced by different positive (d) and negative (e) pulse voltages with a pulse width of 10 ms. (f) SVDP index versus pulse voltage in the Ag/Cs_2SnI_6/ITO device.

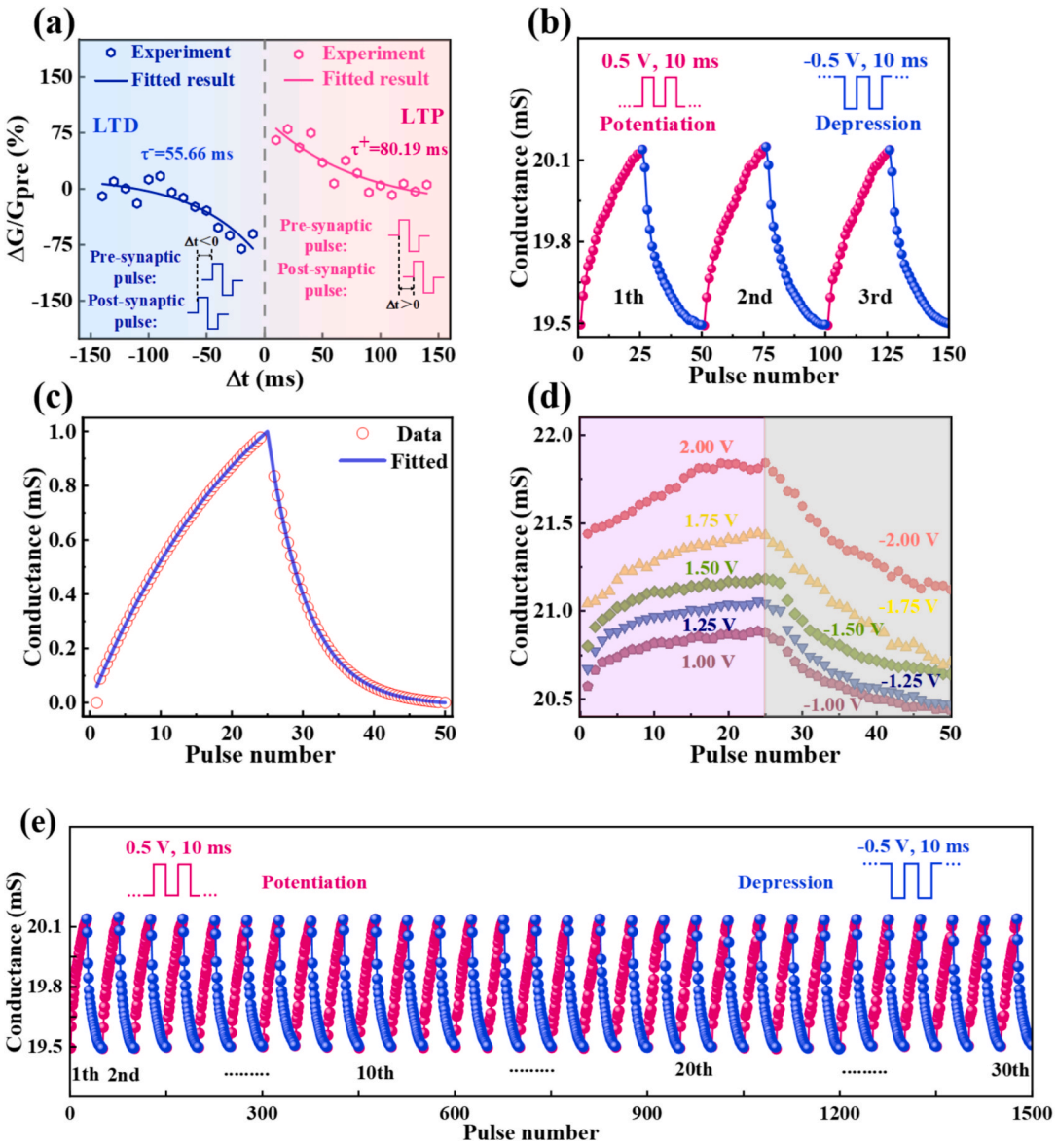


Fig. 5. (a) Variation of $\Delta G/G_{pre}$ of the Ag/Cs_2SnI_6/ITO device with presynaptic/post-synaptic pulse time difference (Δt). (b) Variation of conductance of the Ag/Cs_2SnI_6/ITO device with three consecutive cycles (V_{read} = 0.1 V). (c) Fitting result of the potentiation-depression curve based on the equations (6) and (7), respectively. (d) Conductance of Ag/Cs_2SnI_6/ITO device versus different pulse voltages (V_{read} = 0.1 V). (e) Conductance of the Ag/Cs_2SnI_6/ITO device with thirty consecutive cycles (V_{read} = 0.1 V).

term plasticity [47].

Short-term plasticity is triggered by short-term stimuli. As shown in Fig. 3(b), under the stimulation of a single pulse (0.3/-0.3 V, 10 ms), the conductance of Ag/Cs_2SnI_6/ITO device abruptly rose/dropped to a larger/lower value. It slowly decreased/increased at a reading voltage of 0.1 V. The phenomenon simulated the excitatory/inhibitory post-synaptic current (EPSC/IPSC) behavior of the synapse. Moreover, two consecutive pulse stimuli (0.3/-0.3 V, 10 ms) with a pulse interval time (ΔT) of 10 ms were applied to the device. The conductance response triggered by the second pulse (A_2) was always larger/lower than that of the first pulse (A_1), simulating the PPF/PPD properties of the synapse, representing the cumulative effect of two consecutive presynaptic stimuli on postsynaptic current, as shown in Fig. 3(c). Then, the dependence of the increment of synaptic weights on the pulse interval was analyzed by changing ΔT. The PPF/PPD index can be calculated by the following equation:

$$PPF/PPD\ Index = \frac{A_2 - A_1}{A_1} \times 100\% \quad (1)$$

The calculation results are shown in Fig. 3(d). As ΔT increases, the PPF index decays exponentially and the PPD index increases exponentially, which corresponds to the biological synaptic properties. These results confirm that the prepared Ag/Cs_2SnI_6/ITO devices are able to successfully simulate short-term plasticity, indicating their potential applications in learning memory simulation, artificial neural system development, and so on.

Meanwhile, spike-duration-dependent plasticity (SDDP) and spike-voltage-dependent plasticity (SVDP) are also important parts of synaptic plasticity. SVDP emphasizes the effect of pulse width and voltage amplitude of pre-synaptic or post-synaptic neuron action potential on the regulation of synaptic strength [48,49]. The conductance response results of SDDP and SVDP simulated on this device are shown in Fig. 4. With increasing the pulse width from 0.01 ~ 0.2 s, the conductance of EPSC strength gradually increased and the conductance of IPSC strength

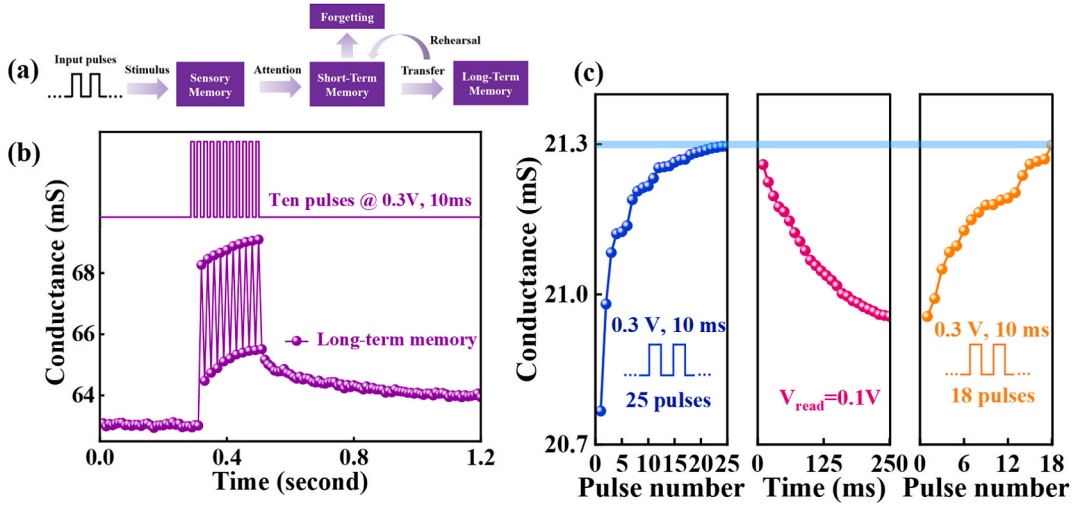


Fig. 6. (a) Schematic of human memory in the Atkinson and Shiffrin model. (b) Conductance response of the Ag/Cs_2SnI_6/ITO device after 10 pulses. (c) Simulation of learning, forgetting and relearning process using the Ag/Cs_2SnI_6/ITO device (V_{read} = 0.1 V).

continually decreased, as shown in Fig. 4(a) and (b), respectively. The relaxation time of the device was estimated over 1.2 s. The duration of above applied pulses is smaller than the relaxation time, ensuring the steady-state excitation [50–52]. With a same pulse width of 10 ms, the dependence of conductance on different pulse voltage stimulation was also studied in the Ag/Cs_2SnI_6/ITO device. From Fig. 4(d) and (e), it is found that the conductance of EPSC strength increased but the conductance of IPSC strength decreased with the increasing the pulse amplitude from 1.0 ~ 2.0 V, which clearly showed the dependence of the memory behavior on the pulse width and amplitude. The SDDP/SVDP index can be quantitatively calculated using the following formula [53,54]:

$$SDDP \Big/ SVDP\ Index = \frac{A'_n - A'_0}{A'_0} \times 100\% \quad (2)$$

A'$_0$ and A'$_n$ are the conductance values for pulses width of 10 and n (n = 50–200) ms/pulses amplitude of 1 and n (n = 1.25–2.00) V. The calculated SDDP index is plotted in Fig. 4(c), showing the value increases (orange circles) and decreases (green circles) with increasing pulse width under 0.3 and -0.3 V, respectively. As drawn in Fig. 4(f), the SVDP index increases with increasing the positive voltage (orange squares) but decreases with reducing the negative voltage (green squares).

Spiking time-dependent plasticity (STDP) is the biological synapse key synaptic function in simulation and refers to the phenomenon that synaptic weights vary with the time interval between pre-synaptic and post-synaptic neuronal responses [55,56]. Long-term plasticity of synapses consists of LTP and LTD [57]. To simulate the STDP behavior in Ag/Cs_2SnI_6/ITO devices, we applied pulse voltages by varying the time interval between pulses (Δt), as shown in Fig. 5(a). When the stimulus applied to the presynaptic membrane precedes the postsynaptic membrane (Δt > 0), the increase in synaptic weights is referred to as LTP, and the relative change in synaptic plasticity ($\Delta G/G_{pre}$) decreases markedly with increasing Δt. In contrast, when the pulse stimulation of the presynaptic membrane is later than the postsynaptic membrane (Δt < 0), the decrease of synaptic weights is referred to as LTD [6,58]. The dependence of $\Delta G/G_{pre}$ on Δt can be fitted by an exponential function [59,60]:

$$\Delta G = G_{post} - G_{pre} \quad (3)$$

$$\frac{\Delta G^+}{G_{pre}} = A^+ \exp\left(-\frac{\Delta t}{\tau^+}\right);\ \Delta t > 0 \quad (4)$$

$$\frac{\Delta G^-}{G_{pre}} = A^- \exp\left(-\frac{\Delta t}{\tau^-}\right);\ \Delta t < 0 \quad (5)$$

Where G_{pre} and G_{post} are represented as the conductance values for the presynaptic and postsynaptic membranes of the Ag/Cs_2SnI_6/ITO device, respectively, A is the amplitude and τ represents the time constant, the fitted results of τ^+ and τ^- values were 80.19 and 55.66 ms, respectively. These values are comparable to the millisecond-scale response times observed in biological synapses, indicating the device's potential for emulating LTP and LTD functionality [60,61]. The synaptic excitatory/inhibitory behavior in a biological organism was simulated by applying consecutive positive (0.5 V, 10 ms) and negative (-0.5 V, 10 ms) voltage pulses to the device, as shown in Fig. 5(b). It is found that the conductance values of the device after 3 consecutive cycles of potentiation (pink dots) and depression (blue dots). The conductance values continuously increase/decrease when the device is stimulated by 25 consecutive positive voltage pulses followed by 25 negative voltage pulses for each cycle.

To further evaluate the neuromorphic computing potential for the present Ag/Cs_2SnI_6/ITO device, the experimental potentiation and depression profiles were theoretically analyzed [62,63], as shown in Fig. 5(c). The following formulas (6) and (7) were used to fit the experimental potentiation and depression, respectively. Here, G^+ and G^- represent normalized synaptic conductance for potentiation and depression, respectively, P indicates applied pulse count, P_{max} defines the maximum pulses per operational phase, and σ represents normally distributed statistical noise capturing intrinsic device variability. ν^+ and ν^- are the nonlinearity coefficients, which quantify synaptic weight update deviations from ideality. A zero-coefficient condition corresponds to the theoretically ideal case of linear conductance modulation. In the present device, the extracted results for ν^+ and ν^- are 0.041 and 0.18 for potentiation and depression, respectively. The small coefficients establish the critical benchmarks for analog synaptic simulation [62].

$$G^+ = G^+_{min} + \frac{G^+_{max} - G^+_{min}}{1 - e^{-\nu^+ \cdot P_{max}}} \cdot \left(1 - e^{-\nu^+ \cdot P}\right) + \sigma \quad (6)$$

$$G^- = G^-_{min} + \frac{G^-_{max} - G^-_{min}}{1 - e^{-\nu^- \cdot P_{max}}} \cdot \left(1 - e^{\nu^- \cdot (P - P_{max})}\right) + \sigma \quad (7)$$

Fig. 5(d) shows the conductance of Ag/Cs_2SnI_6/ITO device response to the various pulse voltage under 50 pulse numbers. The first 25 pulses applied are excitatory voltage stimuli, denoted using the positive voltages. The subsequent 25 pulses are inhibitory voltage stimuli, labeled using the negative voltages. This clearly exhibits the conductance

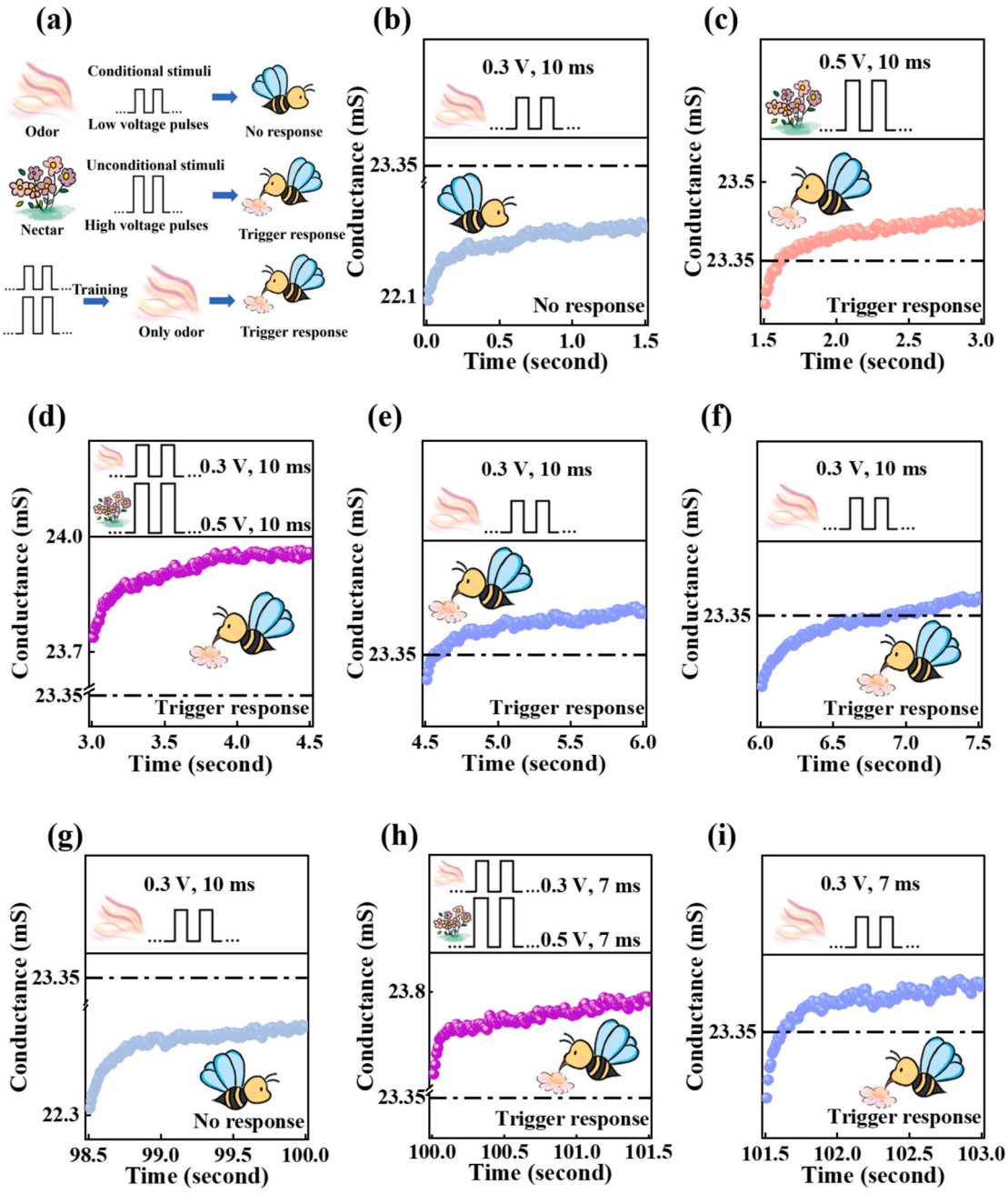


Fig. 7. The PER experiment of bee simulated using the Ag/Cs_2SnI_6/ITO device. (a) Schematic diagram of the PER experiment of bee. The low pulse voltages simulate conditioned stimuli for odor (no trigger response). The high pulse voltages simulate unconditioned stimuli (nectar) corresponding to trigger response. (b) No response to a single odor stimulus under the stimuli voltage (0.3 V, 10 ms). (c) Trigger response to individual nectar stimulation under the stimuli voltage (0.5 V, 10 ms). (d) Trigger response trained by both odor and nectar stimuli, corresponding to the acquisition process. (e) The conductance response with a single odor stimulus after training. (f) A further decay of conductance response to a single odor stimulus but sill remaining the trigger response. (g) The conductance response is below the threshold of 23.35 mS, corresponding to the extinction process. (h) Re-trigger the conductance response by the same pulse voltages but low pulse widths for odor (0.3 V, 7 ms) and nectar (0.5 V, 7 ms) stimulus, achiving the generalization process. (i) The trigger response is kept with a single odor stimulus after re-training.

dependence on the applied voltage for the current Ag/Cs_2SnI_6/ITO device. A higher voltage amplitude causes faster and larger variation of conductance, implying a modulable synaptic plasticity [43]. To evaluate the reproducibility of the device performance, 1500 consecutive voltage pulses were applied to the device. As shown in Fig. 5(e), the device conductance basically keeps the same value under 30 consecutive potentiation and decay cycles, indicating excellent repeatability [63,64]. Therefore, we can simulate the biological associative learning behaviors using the present Cs_2SnI_6 perovskite memristors.

LTM and STM are distinguished based on the time required to return to the resting conductance following synaptic stimulation. The transition from STM to LTM is one of the important synaptic features describing the behavior of memory [6], in accordance with the multi-store model of memory proposed by Atkinson and Shiffrin [65], as shown in Fig. 6(a). The model conceptualizes memory processes as a system, which can be categorized into three distinct subsystems based on the duration of information retention: sensory memory, STM and LTM [65]. This memory behavior was simulated on the Ag/Cs_2SnI_6/ITO device. Fig. 6(b) demonstrates that 10 repeated pulse stimuli (0.3 V, 10 ms) enhance the synaptic weight within a short period. It is clearly

observed that the conductance increases with the number of pulses and subsequently stabilizes at a certain level, without decay after the cessation of the pulses, indicating a transition from STM to LTM. Thus, the device, analogous to biological synapses, exhibits significant potential for future applications. Another notable biological synaptic characteristic exhibited by the Ag/Cs_2SnI_6/ITO memristor is the "learning-forgetting-relearning" process, which is typically described by the Ebbinghaus forgetting curve, demonstrating that human memory is reinforced through learning [66,67]. As depicted in Fig. 6(c), the learning process was initially simulated by applying 25 identical pulse stimuli (0.3 V, 10 ms) to the device. The conductance exhibited a significant increase with the number of voltage pulses, indicating the memorization of information. Subsequently, upon the removal of the pulse stimuli, spontaneous decay of the conductance was observed, simulating the forgetting process of partial information. This decay did not revert to the initial state, suggesting that a portion of the information had been converted into LTM. Finally, the pre-forgetting memory level was restored by applying only 18 identical pulse stimuli (0.3 V, 10 ms), simulating the relearning process. Notably, the number of pulses required for relearning was fewer than that for the initial learning phase. These memory consolidation phenomena show a cumulative memory effect similar to that of the human brain. This implies that relearning forgotten information requires less time compared to the initial learning process, demonstrating the device's capability to rapidly emulate the memory functions of the human brain [68–71]. Thus, the device can achieve the learning and memory functions, which is the precondition for subsequent study on the associative learning behaviors.

As a typical classic conditioned reflex experiment, the PER behavior of animals could show the associative learning behaviors of the brain. The synaptic hardware with associative learning capabilities has demonstrated significant application potential in fields such as brain-machine interfaces and neural control [72,73]. The acquisition, extinction, and generalization are three fundamental characteristics of classical conditioning, corresponding to the storage of information, the elimination of old information, and the storage of new information in the biological brain [9]. Here, the PER behavior of bees was tested using the present Ag/Cs_2SnI_6/ITO devices. The mechanism was illustrated in Fig. 7(a). Low-voltage pulse and high-voltage pulse applied on the device represent the odor signal and nectar signal, respectively. Both low and high voltage pulses simultaneously applied to the device, meaning the odor and nectar signals appear concurrently. As shown in Fig. 7(b), the low-voltage pulses with 0.3 V and 10 ms applied on the device represent the odor signal. The conductance value of the device was set at 23.35 mS as the threshold for PER, where the conductance value exceeding the threshold signifies the PER behavior of bees. Initially, the conductance value is below the threshold, indicating that the reflex was not triggered, and the bees did not exhibit the PER behavior. The high-voltage pulses with 0.5 V and 10 ms applied on the device denote the nectar signal, as drawn in Fig. 7(c), which caused the conductance to exceed the threshold of 23.35 mS. As a result, the bees showed the PER behavior. Subsequently, the above low and high voltage pulses simultaneously applied to the device, leading to a significant increase in conductance that surpassed the proboscis extension threshold, which corresponded to the training process (Fig. 7(d)). After training, the PER behavior of bees could also be triggered even if only the low voltage pulses were applied, as shown in Fig. 7(e), establishing an association between the unconditioned and conditioned reflexes, that is, acquisition. If the training was stopped, the conductance gradually decayed with time (Fig. 7(f)) until the conductance is below the threshold (Fig. 7(g)). This extinction indicated that the conditioned reflex would gradually disappear and the associative learning behavior is forgotten by the bee. Meanwhile, the generalization process was also studied in Fig. 7(h) and (i), which can trigger the same conditioned reflex by similar stimulus. The low-voltage pulses (0.3 V, 7 ms) and high-voltage pulses (0.5 V, 7 ms) were used as the similar stimulus and simultaneously applied to the device. As shown in Fig. 7(h), the conductance exceeded the threshold of 23.35 mS, effectively re-triggering the PER behavior, result in the generalization. This can be remained after withdrawing the high-voltage pulses (0.5 V, 7 ms) in Fig. 7(i). It is noted that the decrease of pulse width from 10 to 7 ms reduces the conductance by comparing the results of Fig. 7(d) and (h), agreeing with the observation of Fig. 4. Based on the experimental results, the Ag/Cs_2SnI_6/ITO device successfully simulates the PER behavior of bees, establishing the associative learning behavior.

4. Conclusion

The vertical memristor with Ag/Cs_2SnI_6/ITO structure was prepared, exhibiting excellent RS behavior and successfully simulating both short-term and long-term plasticity of biological synapses. Moreover, the Ag/Cs_2SnI_6/ITO device simulates the cognitive process of "learning-forgetting-relearning" in human brain, establishes associative learning and memory behavior by simulating the PER behavior of bees with three characteristics of classical conditioning. This study opens up a new way to develop dual-function devices with non-volatile information storage and artificial synaptic functions.

CRediT authorship contribution statement

Jiaxin Gao: Writing – original draft, Software, Resources, Methodology, Investigation, Data curation, Conceptualization. **Lingyu Meng:** Formal analysis. **Hanghui Li:** Data curation. **Feifei Luo:** Software. **Fei Wang:** Software. **Xianmin Zhang:** Writing – review & editing.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at https://doi.org/10.1016/j.cej.2025.163362.

Data availability

Data will be made available on request.

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